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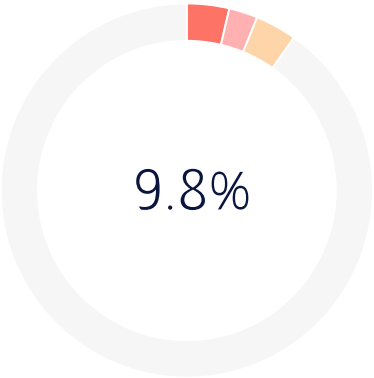

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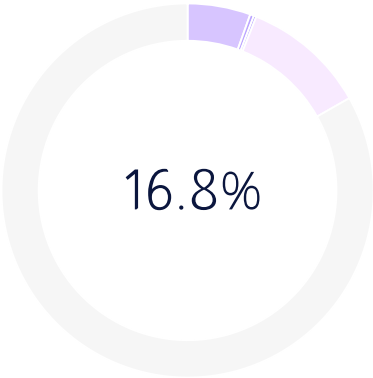

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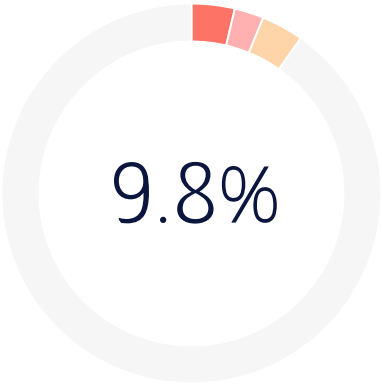

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A Study of E-Learning Systems and Their Impact on Student Performance

Sachin Singh
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
sachin2203.be22@chitkara.edu.in

Ronit Sharma
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
ronit2185.be22@chitkara.edu.in

Aryan Rathor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
aryan1385.be22@chitkara.edu.in

Porush Yadav
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
porush2047.be22@chitkara.edu.in

Corresponding Author
Dr. Shikha Tuteja
Associate Professor
Department of Computer Science & Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India
shikha.1290@chitkara.edu.in

***Abstract*—In recent years e-learning has become very important part of our education system. As e-learning system is helping students to attend classes even in pandemic situations like COVID-19.They are improving without any difficulties because they are getting proper educational support without even if they are not able to attend the classes.With the help of e-learning systems students can study from anywhere in the world they just need internet access and nothing else.There are many e-learning platforms like vedantu, unacademy etc. that are providing the students with live and recorded video lectures, they can download study material, take assignments and even get in contact with the teachers themselves to clear any doubts they have because of 24X7 support available on these platforms.**

We are doing this study to understand how exactly e-learning systems impact students performance.Using this study/survey our main focus is to find whether students are able to learn better, stay consistent and more motivated and achieving high marks after using e-learning systems or not. We will also tell you advantages and disadvantages students are having using e-learning systems.

We basically used mix-method approach for this survey of e-learning systems and their impacts on students performance and achievements. We got our primary data from questionnaire we made using google forms and distributed these questionnaire with college students and school students. We designed questions in such a way that we can understand students level of satisfaction with the system, their experience, and challenges they faced while studying online. Secondary data was collected from reviewing few similar kind of research paper studies on e-learning systems.

This study shows several benefit of studying online like flexible timings, access to a lot of study materials at anytime from anywhere.Students can learn easily with their own pace, can watch videos on repeat if they do not understand the topic. However survey also reported that students find few difficulties while studying online like internet not stable enough, if weather is bad and electricity is gone no availability of devices. They also lack direct interaction with teachers, sometimes find it hard to focus, and some of them experience technical problems with platform.

Overall our study shows that e-learning system would not be able to take over the traditional way of teaching at school.It is like a supporting system that students can use side by side with regular classes.If implemented better e-learning system can improve the flexibility for students.This study shows useful information about how colleges and other e-learning systems can improve and create better system to help out students more effectively.

***Index Terms*—E-learning, online learning, student performance, academic achievement, student engagement, digital education, educational technology, blended learning**

I. Introduction

A. Background of E-Learning

The phrase "e-learning" entered formal academic and professional discourse around 1999, when it was introduced at a major technology and learning conference in the United States. That said, the use of digital tools for educational purposes stretches back several decades before the term was coined. As far back as the 1960s, university researchers and professors began experimenting with computers as instructional aids — using them to teach foundational subjects like arithmetic and spelling to young students. Around the same time, institutions such as the University of Illinois began building internal digital networks that allowed students to retrieve course content and revisit recorded lectures from shared terminals. This early experimentation laid the groundwork for what would eventually grow into the modern e-learning ecosystem. By the mid-1980s, several universities had extended this access further,

enabling students to retrieve academic materials from library-based computer terminals.

B. Evolution of E-Learning Systems

E-learning, in its earliest form, emerged as a computer-assisted method of instruction in the 1960s, but it did not gain widespread adoption until the internet became publicly accessible. Over the following decades, the tools and techniques used for digital education transformed significantly — from simple text-based programs to rich, interactive multimedia platforms. What was once limited to institutional networks has now expanded to cover education, corporate training, military instruction, and self-directed learning globally.

As technology matured, so did the variety of approaches to online learning. Instructional designers and developers began incorporating features like adaptive content delivery, collaborative tools, and real-time feedback into their platforms. Despite this rapid technological growth, researchers have noted that evidence on the actual effectiveness of different instructional methods within e-learning remains limited and inconsistent. Identifying which pedagogical strategies truly improve student outcomes continues to be an open question in the field.

Among the most notable recent shifts in e-learning are the rise of blended learning models that combine face-to-face and digital instruction, the growth of micro-learning which delivers content in short focused sessions, personalized learning pathways driven by student data, and the global expansion of Massive Open Online Courses (MOOCs) that make quality education accessible regardless of geography. These developments continue to define the future direction of educational technology.

C. Importance in Modern Education

The significance of e-learning in modern education became especially visible during the COVID-19 pandemic, which forced schools and universities worldwide to shift to remote instruction almost overnight. What had previously been a supplement to traditional teaching suddenly became the primary mode of education for millions of students. This rapid transition demonstrated both the potential and the real challenges of digital learning at scale.

Today, schools and universities increasingly rely on e-learning systems not just as emergency alternatives but as permanent components of their educational offerings. Digital platforms provide students with the flexibility to learn at their own pace, access diverse resources, and engage with content outside the fixed hours of a traditional classroom. Tools such as video conferencing software, learning management systems, mobile applications, and cloud-based content repositories have all become part of the everyday learning experience. This shift has encouraged educators and institutions to rethink how quality education is delivered, moving away from rigid one-size-fits-all models toward more adaptable, technology-supported approaches.

D. Purpose of the Study

Given the foregoing, this study was carried out to; (1) investigate the online learning experiences of students, (2) review the academic performance of students using online learning platforms, and (3) explore the levels of students' engagement when learning using online platforms.

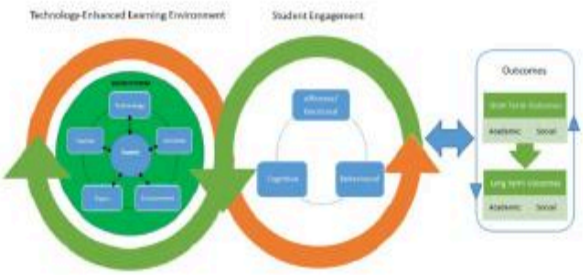

Fig. 1. Conceptual Model of E-Learning System Impact on Student Performance

II. Literature Review

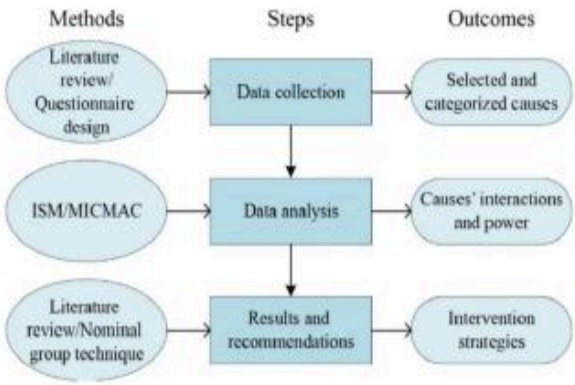

Fig. 2. Research Framework of the Study

III. Proposed Methodology

A. Research Design

In this study we wanted to evaluate that how do e-learning systems actually affect student performance. To carry this out effectively , we decided to perform a mixed-method approach. This means we collected both numerical (quantitative data) and personal opinions (qualitative data) from students . Numbers alone cannot tell us how students actually feel while practicing e-learning , so this approached seemed to give a more complete picture .

Our main objective was to know the real advantages and issues students go through while e-learning. We focused on things like flexibility , easy access to materials , student engagement , satisfaction , and overall academic performance

B. Data Collection

To gather primary data, we used simple questionnaires through google forms [4]. We created around 5 questions that were easy to answer. Some of the questions which we asked included asking students how often they use e-learning platforms, how comfortable they usually feel when they study online, what problems they go through, whether they see improvement or decrement in their marks after shifting to online classes.

We shared this google form with students of our university along with a few school students via wats app groups and college email. We received many responses which helped us evaluate the real experiences of the students.

C. Data Sources

Other than the survey , we also gathered secondary data . We went through many research papers , journal articles , and reports which were available on the internet about e-learning and student performance . This secondary data which we gathered , helped us collect background knowledge and helped us in comparison between our survey results and what other researchers have already found [2] [3].

D. Data Analysis

After gathering all the responses we analyzed the collected data in a very simple way. We went through the percentages of answers for each question. We also tried to look for similar patterns in what students wrote in the open-ended questions. for example many of the students mentioned similar problems like bad internet, and lack of teacher-students interaction. This whole analysis helped us understand the overall thought process and opinions of the students about e-learning systems.

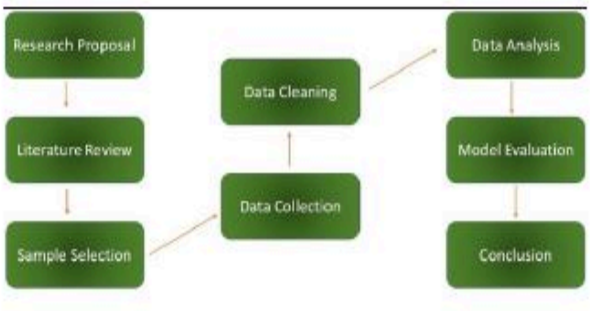

Fig. 3. Research Methodology Process

IV. Proposed Model

In this section, e present a very simple model to help explain how e-leaning systems are affecting students performance. We think that e-learning does not work by itself. The success of e-learning depends heavily on several factors working together.

A. E-learning System Factors

The success behind online learning depends on alot of things, the most important factors are

1. Ease of use of the platform.
2. If students are getting easy access to study materials.
3. If contents quality is good or not.
4. What are the interactions medium.

If the platform is very complex to use for normal students or it is slow and does not work fast, students does not prefer using such platform. And if platform is fast and simple to use for students and UI is simple and clear and site loads fast students prefer using it because of how beginner friendly it is [3].

B. Student Engagement

Students engagement is very important part of learning and understanding. It basically means if student are actively participating giving tests and assignments on time or not. If they watch lectures and videos carefully take notes of everything they learn or not. If they do not understand something they ask their doubts to mentors or teachers.

In online learning engagement depends heavily on unique features the plaform provides. For example, fun and short video lectures, quizzes and study materials like notes for all the chapters and so on. If the content is boring and there is not interesting thing for students they lose their interest very easily [6].

C. Academic Performance Signs

Students performance with online learning can be seen through very simple Signs.

1. Number of assignments he has completed.
2. Their scores in the online quizzes and test assessments.
3. Their active participation and interaction with teacher on platform.
4. Their marks in each subject that they are learning online.

D. Relationship Between E-learning and Performance

Our study shows displays that online learning systems impact students performance and grades by two things. First one is good and easy to use platform or website like easy to use, best content helps in students to engage more and be active. Second one is when they actively participate their marks automatically increases. In very simple words:

Good e-learning platform -¿ More students participation -¿ Good academic results.

We think that when e-learning platform gives easy access to the students for lectures, study material and useful content, students stay active and motivated and they learn more fast and better.But on the other hand if any of these things are not there students performance decreases [1].

Fig. 4. Conceptual Model of E-Learning System Impact on Student Performance

V. Survey Analysis

We also gathered variety of feedback from the college and school students using google form questionnaire we made [4]. We made all these questions to understand how students actually feel and experiences while learning online and what difficulties they face.

Table we have below is showing how much negative, positive and neutral responses we got:

TABLE I
Student Responses on E-Learning System Usage

S.No	Evaluation Factor	Positive (%)	Neutral (%)	Negative (%)
1	Accessibility of learning materials	78	12	10
2	Flexibility in learning timing	72	15	13
3	Interaction with teachers	65	18	17
4	Use of videos and lectures	70	16	14
5	Satisfaction with e-learning platforms	68	20	12
6	Technical issues during online learning	40	22	38
7	Students participation during online classes	62	21	17
8	Effect on academic performance	66	19	15

From this study, we can see clear results. Most of the students gave positive feedback about accessibility of learning materials (78%) and flexibility in learning timings (72%). Many students also said that they like the use of videos and lectures(70%), saying it made their learning more interesting and easy [7].

On the other hand, there were some weak areas. Like technical issues got the lowest positive response (only 40%), with 38% students facing difficulties like slow internet, or login difficulties. Interaction with teachers was another area of concern — only 65% students were satisfied with it. Many students said that they dont do doubt-clearing sessions with teachers [5].

When we asked about the effect on academic performance, 66% students felt that e-learning had a positive impact on their studies, while 15% felt it had a negative effect. Student participation during online classes was average, with 62% students saying they stay active, but 17% found it hard to focus for long hours on screen [6].

So the survey says that students appreciate the flexibility and convenience of e-learning systems. But technical problems and lack of personal interaction are the big problems they face. This analysis helped us understand both the strengths and weaknesses of online learning from a students point of view [1] [2].

VI. Conclusion

This study shows the role of online learning systems and their effect on students academic performances and marks they are getting. With the sudden growth of digital technologies like internet computer and smartphones, online learning platforms are now becoming important for academic performances of students [1]. The study said many factors like easy accessibility of study materials and flexibility of study timings and active participation in online quizzes and assessment are helping students alot.

Our studys result shows that e-learning systems generally have very good impact on students marks and performance when platform is fast, and simple to understand and filled with interesting activities [2] [3]. Students are getting many benifits like recorded lectures, free content and many more things that help them to boost their performance easily and effectively. But on the other hand students are also facing problems like slow internet, site crashes and no technical support on the platforms [6].

Overall e-learning systems can be used as complementary approach to tradition classroom learning instead of complete replacement. When it is used effectively, it can increase students performance and support for better academic result [7].

Fig. 5. Overall Impact of E-Learning Systems on Student Performance

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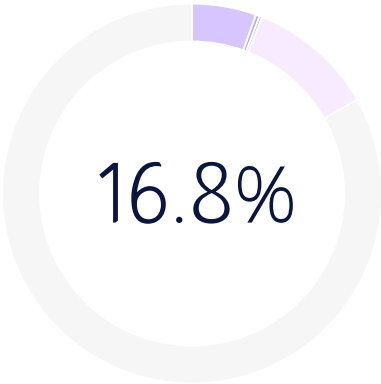

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Omitted Words		0

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AI Phrases

Detects phrases that appear with higher frequency in AI-written text than in human writing.

The frequency of a phrase in AI vs. human text.

4 x — 1254x

1254x **Tools such as video conferencing software,**

How frequently the phrase was found in our dataset:

AI Text	**4.82 / 1,000,000 Documents**
Human Text	**0 / 1,000,000 Documents**

438x **students with the flexibility to learn at their own**

How frequently the phrase was found in our dataset:

AI Text	**1.68 / 1,000,000 Documents**
Human Text	**0 / 1,000,000 Documents**

205x **blended learning models that combine**

How frequently the phrase was found in our dataset:

AI Text	**3.41 / 1,000,000 Documents**
Human Text	**0.02 / 1,000,000 Documents**

170x **incorporating features like**

How frequently the phrase was found in our dataset:

AI Text	**41.1 / 1,000,000 Documents**
Human Text	**0.24 / 1,000,000 Documents**

169x **models toward more**

How frequently the phrase was found in our dataset:

AI Text	**23.41 / 1,000,000 Documents**
Human Text	**0.14 / 1,000,000 Documents**

111x **experience. This shift has**

How frequently the phrase was found in our dataset:

AI Text	**0.99 / 1,000,000 Documents**
Human Text	**0.01 / 1,000,000 Documents**

107x **Importance in Modern Education**

How frequently the phrase was found in our dataset:

AI Text	**1.09 / 1,000,000 Documents**
Human Text	**0.01 / 1,000,000 Documents**

89x **personalized learning pathways**

How frequently the phrase was found in our dataset:

AI Text	**40.04 / 1,000,000 Documents**
Human Text	**0.45 / 1,000,000 Documents**

45x **in short focused sessions,**

How frequently the phrase was found in our dataset:

AI Text	**1.5 / 1,000,000 Documents**
Human Text	**0.03 / 1,000,000 Documents**

45x **online learning platforms, and**

How frequently the phrase was found in our dataset:

AI Text	**47.85 / 1,000,000 Documents**
Human Text	**1.07 / 1,000,000 Documents**

30x **and engage with content**

How frequently the phrase was found in our dataset:

AI Text	**19.94 / 1,000,000 Documents**
Human Text	**0.66 / 1,000,000 Documents**

22x **moving away from rigid**

How frequently the phrase was found in our dataset:

AI Text	**3.83 / 1,000,000 Documents**
Human Text	**0.17 / 1,000,000 Documents**

22x **pandemic, which forced**

How frequently the phrase was found in our dataset:

AI Text	**8.32 / 1,000,000 Documents**
Human Text	**0.38 / 1,000,000 Documents**

17x **both the potential and the**

How frequently the phrase was found in our dataset:

AI Text	**31.54 / 1,000,000 Documents**
Human Text	**1.87 / 1,000,000 Documents**

14x **educators and institutions to**

How frequently the phrase was found in our dataset:

AI Text	**4.33 / 1,000,000 Documents**
Human Text	**0.31 / 1,000,000 Documents**

14x **this rapid technological**

How frequently the phrase was found in our dataset:

AI Text	**3.61 / 1,000,000 Documents**
Human Text	**0.27 / 1,000,000 Documents**

13x **universities increasingly rely**

How frequently the phrase was found in our dataset:

AI Text	**1.01 / 1,000,000 Documents**
Human Text	**0.08 / 1,000,000 Documents**

13x **adaptive content delivery,**

How frequently the phrase was found in our dataset:

AI Text	**2.41 / 1,000,000 Documents**
Human Text	**0.18 / 1,000,000 Documents**

13x **These developments continue to**

How frequently the phrase was found in our dataset:

AI Text	**2.2 / 1,000,000 Documents**
Human Text	**0.17 / 1,000,000 Documents**

13x **feedback into their**

How frequently the phrase was found in our dataset:

AI Text	**12.77 / 1,000,000 Documents**
Human Text	**0.99 / 1,000,000 Documents**

13x **effectiveness of different instructional**

How frequently the phrase was found in our dataset:

AI Text	**1.42 / 1,000,000 Documents**
Human Text	**0.11 / 1,000,000 Documents**

13x **so did the variety of**

How frequently the phrase was found in our dataset:

AI Text	**1.58 / 1,000,000 Documents**
Human Text	**0.13 / 1,000,000 Documents**

11x **their educational offerings.**

How frequently the phrase was found in our dataset:

AI Text	**6.16 / 1,000,000 Documents**
Human Text	**0.54 / 1,000,000 Documents**

10x **supplement to traditional teaching**

How frequently the phrase was found in our dataset:

AI Text	**1.32 / 1,000,000 Documents**
Human Text	**0.13 / 1,000,000 Documents**

10x **schools and universities worldwide**

How frequently the phrase was found in our dataset:

AI Text	**4.66 / 1,000,000 Documents**
Human Text	**0.48 / 1,000,000 Documents**

9x **quality education accessible**

How frequently the phrase was found in our dataset:

AI Text	**9.92 / 1,000,000 Documents**
Human Text	**1.12 / 1,000,000 Documents**

9x **System Impact on**

How frequently the phrase was found in our dataset:

AI Text	**17.19 / 1,000,000 Documents**
Human Text	**2.01 / 1,000,000 Documents**

8x **significantly — from simple**

How frequently the phrase was found in our dataset:

AI Text	**2.44 / 1,000,000 Documents**
Human Text	**0.32 / 1,000,000 Documents**

7x **systems, mobile applications, and**

How frequently the phrase was found in our dataset:

AI Text	**1.27 / 1,000,000 Documents**
Human Text	**0.17 / 1,000,000 Documents**

7x **education for millions of students.**

How frequently the phrase was found in our dataset:

AI Text	**1.14 / 1,000,000 Documents**
Human Text	**0.16 / 1,000,000 Documents**

7x **became the primary mode of**

How frequently the phrase was found in our dataset:

AI Text	**1.28 / 1,000,000 Documents**
Human Text	**0.19 / 1,000,000 Documents**

7x **limited to institutional**

How frequently the phrase was found in our dataset:

AI Text	**4.67 / 1,000,000 Documents**
Human Text	**0.71 / 1,000,000 Documents**

6x **learning experiences of students,**

How frequently the phrase was found in our dataset:

AI Text	**7.13 / 1,000,000 Documents**
Human Text	**1.23 / 1,000,000 Documents**

5x **This rapid transition**

How frequently the phrase was found in our dataset:

AI Text	**3.24 / 1,000,000 Documents**
Human Text	**0.59 / 1,000,000 Documents**

5x **of Massive Open Online Courses (MOOCs)**

How frequently the phrase was found in our dataset:

AI Text	**11 / 1,000,000 Documents**
Human Text	**2.02 / 1,000,000 Documents**

5x **did not gain widespread**

How frequently the phrase was found in our dataset:

AI Text	**4.92 / 1,000,000 Documents**
Human Text	**0.95 / 1,000,000 Documents**

5x **the tools and techniques used for**

How frequently the phrase was found in our dataset:

AI Text	**1.22 / 1,000,000 Documents**
Human Text	**0.26 / 1,000,000 Documents**

4x **approaches to online**

How frequently the phrase was found in our dataset:

AI Text	**8.03 / 1,000,000 Documents**
Human Text	**1.8 / 1,000,000 Documents**

4x **an open question in the field.**

How frequently the phrase was found in our dataset:

AI Text	**1.24 / 1,000,000 Documents**
Human Text	**0.28 / 1,000,000 Documents**

4x **the Study Given the**

How frequently the phrase was found in our dataset:

AI Text	**7.83 / 1,000,000 Documents**
Human Text	**2.04 / 1,000,000 Documents**

A Study of E-Learning Systems and Their Impact on Student Performance

Sachin Singh
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
sachin2203.be22@chitkara.edu.in

Ronit Sharma
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
ronit2185.be22@chitkara.edu.in

Aryan Rathor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
aryan1385.be22@chitkara.edu.in

Porush Yadav
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology
porush2047.be22@chitkara.edu.in

Corresponding Author
Dr. Shikha Tuteja
Associate Professor
Department of Computer Science & Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India
shikha.1290@chitkara.edu.in

***Abstract*—In recent years e-learning has become very important part of our education system. As e-learning system is helping students to attend classes even in pandemic situations like COVID-19.They are improving without any difficulties because they are getting proper educational support without even if they are not able to attend the classes.With the help of e-learning systems students can study from anywhere in the world they just need internet access and nothing else.There are many e-learning platforms like vedantu, unacademy etc. that are providing the students with live and recorded video lectures, they can download study material, take assignments and even get in contact with the teachers themselves to clear any doubts they have because of 24X7 support available on these platforms.**

We are doing this study to understand how exactly e-learning systems impact students performance.Using this study/survey our main focus is to find whether students are able to learn better, stay consistent and more motivated and achieving high marks after using e-learning systems or not. We will also tell you advantages and disadvantages students are having using e-learning systems.

We basically used mix-method approach for this survey of e-learning systems and their impacts on students performance and achievements. We got our primary data from questionnaire we made using google forms and distributed these questionnaire with college students and school students. We designed questions in such a way that we can understand students level of satisfaction with the system, their experience, and challenges they faced while studying online. Secondary data was collected from reviewing few similar kind of research paper studies on e-learning systems.

This study shows several benefit of studying online like flexible timings, access to a lot of study materials at anytime from anywhere.Students can learn easily with their own pace, can watch videos on repeat if they do not understand the topic. However survey also reported that students find few difficulties while studying online like internet not stable enough, if weather is bad and electricity is gone no availability of devices. They also lack direct interaction with teachers, sometimes find it hard to focus, and some of them experience technical problems with platform.

Overall our study shows that e-learning system would not be able to take over the traditional way of teaching at school.It is like a supporting system that students can use side by side with regular classes.If implemented better e-learning system can improve the flexibility for students.This study shows useful information about how colleges and other e-learning systems can improve and create better system to help out students more effectively.

***Index Terms*—E-learning, online learning, student performance, academic achievement, student engagement, digital education, educational technology, blended learning**

I. Introduction

A. Background of E-Learning

The phrase "e-learning" entered formal academic and professional discourse around 1999, when it was introduced at a major technology and learning conference in the United States. That said, the use of digital tools for educational purposes stretches back several decades before the term was coined. As far back as the 1960s, university researchers and professors began experimenting with computers as instructional aids — using them to teach foundational subjects like arithmetic and spelling to young students. Around the same time, institutions such as the University of Illinois began building internal digital networks that allowed students to retrieve course content and revisit recorded lectures from shared terminals. This early experimentation laid the groundwork for what would eventually grow into the modern e-learning ecosystem. By the mid-1980s, several universities had extended this access further,

enabling students to retrieve academic materials from library-based computer terminals.

B. Evolution of E-Learning Systems

E-learning, in its earliest form, emerged as a computer-assisted method of instruction in the 1960s, but it did not gain widespread adoption until the internet became publicly accessible. Over the following decades, the tools and techniques used for digital education transformed significantly — from simple text-based programs to rich, interactive multimedia platforms. What was once limited to institutional networks has now expanded to cover education, corporate training, military instruction, and self-directed learning globally.

As technology matured, so did the variety of approaches to online learning. Instructional designers and developers began incorporating features like adaptive content delivery, collaborative tools, and real-time feedback into their platforms. Despite this rapid technological growth, researchers have noted that evidence on the actual effectiveness of different instructional methods within e-learning remains limited and inconsistent. Identifying which pedagogical strategies truly improve student outcomes continues to be an open question in the field.

Among the most notable recent shifts in e-learning are the rise of blended learning models that combine face-to-face and digital instruction, the growth of micro-learning which delivers content in short focused sessions, personalized learning pathways driven by student data, and the global expansion of Massive Open Online Courses (MOOCs) that make quality education accessible regardless of geography. These developments continue to define the future direction of educational technology.

C. Importance in Modern Education

The significance of e-learning in modern education became especially visible during the COVID-19 pandemic, which forced schools and universities worldwide to shift to remote instruction almost overnight. What had previously been a supplement to traditional teaching suddenly became the primary mode of education for millions of students. This rapid transition demonstrated both the potential and the real challenges of digital learning at scale.

Today, schools and universities increasingly rely on e-learning systems not just as emergency alternatives but as permanent components of their educational offerings. Digital platforms provide students with the flexibility to learn at their own pace, access diverse resources, and engage with content outside the fixed hours of a traditional classroom. Tools such as video conferencing software, learning management systems, mobile applications, and cloud-based content repositories have all become part of the everyday learning experience. This shift has encouraged educators and institutions to rethink how quality education is delivered, moving away from rigid one-size-fits-all models toward more adaptable, technology-supported approaches.

D. Purpose of the Study

Given the foregoing, this study was carried out to; (1) investigate the online learning experiences of students, (2) review the academic performance of students using online learning platforms, and (3) explore the levels of students' engagement when learning using online platforms.

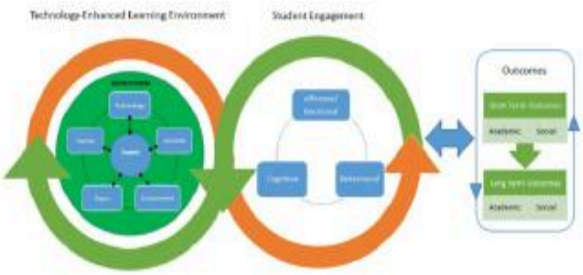

Fig. 1. Conceptual Model of E-Learning System Impact on Student Performance

II. Literature Review

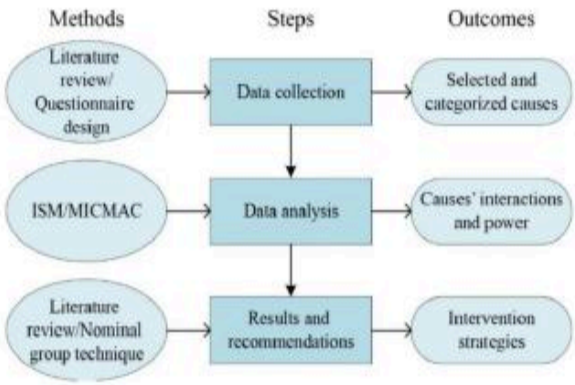

Fig. 2. Research Framework of the Study

III. Proposed Methodology

A. Research Design

In this study we wanted to evaluate that how do e-learning systems actually affect student performance. To carry this out effectively , we decided to perform a mixed-method approach. This means we collected both numerical (quantitative data) and personal opinions (qualitative data) from students . Numbers alone cannot tell us how students actually feel while practicing e-learning , so this approached seemed to give a more complete picture .

Our main objective was to know the real advantages and issues students go through while e-learning. We focused on things like flexibility , easy access to materials , student engagement , satisfaction , and overall academic performance

B. Data Collection

To gather primary data, we used simple questionnaires through google forms [4]. We created around 5 questions that were easy to answer. Some of the questions which we asked included asking students how often they use e-learning platforms, how comfortable they usually feel when they study online, what problems they go through, whether they see improvement or decrement in their marks after shifting to online classes.

We shared this google form with students of our university along with a few school students via wats app groups and college email. We received many responses which helped us evaluate the real experiences of the students.

C. Data Sources

Other than the survey , we also gathered secondary data . We went through many research papers , journal articles , and reports which were available on the internet about e-learning and student performance . This secondary data which we gathered , helped us collect background knowledge and helped us in comparison between our survey results and what other researchers have already found [2] [3].

D. Data Analysis

After gathering all the responses we analyzed the collected data in a very simple way. We went through the percentages of answers for each question. We also tried to look for similar patterns in what students wrote in the open-ended questions. for example many of the students mentioned similar problems like bad internet, and lack of teacher-students interaction. This whole analysis helped us understand the overall thought process and opinions of the students about e-learning systems.

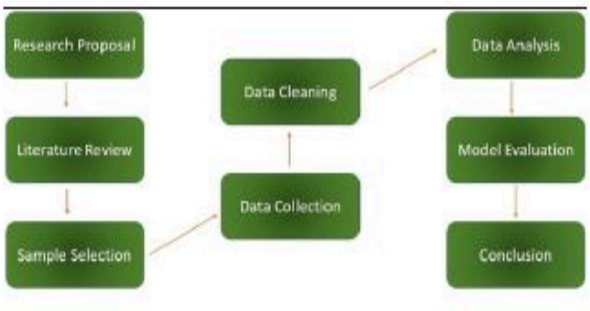

Fig. 3. Research Methodology Process

IV. Proposed Model

In this section, e present a very simple model to help explain how e-leaning systems are affecting students performance. We think that e-learning does not work by itself. The success of e-learning depends heavily on several factors working together.

A. E-learning System Factors

The success behind online learning depends on alot of things, the most important factors are

1. Ease of use of the platform.
2. If students are getting easy access to study materials.
3. If contents quality is good or not.
4. What are the interactions medium.

If the platform is very complex to use for normal students or it is slow and does not work fast, students does not prefer using such platform. And if platform is fast and simple to use for students and UI is simple and clear and site loads fast students prefer using it because of how beginner friendly it is [3].

B. Student Engagement

Students engagement is very important part of learning and understanding. It basically means if student are actively participating giving tests and assignments on time or not. If they watch lectures and videos carefully take notes of everything they learn or not. If they do not understand something they ask their doubts to mentors or teachers.

In online learning engagement depends heavily on unique features the plaform provides. For example, fun and short video lectures, quizzes and study materials like notes for all the chapters and so on. If the content is boring and there is not interesting thing for students they lose their interest very easily [6].

C. Academic Performance Signs

Students performance with online learning can be seen through very simple Signs.

1. Number of assignments he has completed.
2. Their scores in the online quizzes and test assessments.
3. Their active participation and interaction with teacher on platform.
4. Their marks in each subject that they are learning online.

D. Relationship Between E-learning and Performance

Our study shows displays that online learning systems impact students performance and grades by two things. First one is good and easy to use platform or website like easy to use, best content helps in students to engage more and be active. Second one is when they actively participate their marks automatically increases. In very simple words:

Good e-learning platform -¿ More students participation -¿ Good academic results.

We think that when e-learning platform gives easy access to the students for lectures, study material and useful content, students stay active and motivated and they learn more fast and better.But on the other hand if any of these things are not there students performance decreases [1].

Fig. 4. Conceptual Model of E-Learning System Impact on Student Performance

V. Survey Analysis

We also gathered variety of feedback from the college and school students using google form questionnaire we made [4]. We made all these questions to understand how students actually feel and experiences while learning online and what difficulties they face.

Table we have below is showing how much negative, positive and neutral responses we got:

TABLE I
Student Responses on E-Learning System Usage

S.No	Evaluation Factor	Positive (%)	Neutral (%)	Negative (%)
1	Accessibility of learning materials	78	12	10
2	Flexibility in learning timing	72	15	13
3	Interaction with teachers	65	18	17
4	Use of videos and lectures	70	16	14
5	Satisfaction with e-learning platforms	68	20	12
6	Technical issues during online learning	40	22	38
7	Students participation during online classes	62	21	17
8	Effect on academic performance	66	19	15

From this study, we can see clear results. Most of the students gave positive feedback about accessibility of learning materials (78%) and flexibility in learning timings (72%). Many students also said that they like the use of videos and lectures(70%), saying it made their learning more interesting and easy [7].

On the other hand, there were some weak areas. Like technical issues got the lowest positive response (only 40%), with 38% students facing difficulties like slow internet, or login difficulties. Interaction with teachers was another area of concern — only 65% students were satisfied with it. Many students said that they dont do doubt-clearing sessions with teachers [5].

When we asked about the effect on academic performance, 66% students felt that e-learning had a positive impact on their studies, while 15% felt it had a negative effect. Student participation during online classes was average, with 62% students saying they stay active, but 17% found it hard to focus for long hours on screen [6].

So the survey says that students appreciate the flexibility and convenience of e-learning systems. But technical problems and lack of personal interaction are the big problems they face. This analysis helped us understand both the strengths and weaknesses of online learning from a students point of view [1] [2].

VI. Conclusion

This study shows the role of online learning systems and their effect on students academic performances and marks they are getting. With the sudden growth of digital technologies like internet computer and smartphones, online learning platforms are now becoming important for academic performances of students [1]. The study said many factors like easy accessibility of study materials and flexibility of study timings and active participation in online quizzes and assessment are helping students alot.

Our studys result shows that e-learning systems generally have very good impact on students marks and performance when platform is fast, and simple to understand and filled with interesting activities [2] [3]. Students are getting many benifits like recorded lectures, free content and many more things that help them to boost their performance easily and effectively. But on the other hand students are also facing problems like slow internet, site crashes and no technical support on the platforms [6].

Overall e-learning systems can be used as complementary approach to tradition classroom learning instead of complete replacement. When it is used effectively, it can increase students performance and support for better academic result [7].

Fig. 5. Overall Impact of E-Learning Systems on Student Performance

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